

Linux module: 6

Assignment

Que: Use ifconfig or IP to view and configure network interfaces.

Ans:

ifconfig

This command can be used to configure or display network interface configuration information, or to assign an address to a network interface. It's also used to define the network address of each interface at system startup. However, the IP command has since replaced ifconfig.

IP

This command has many options, including finding your Linux IP address.

Que: Use ping to test network connectivity.

Ans:

- Open a Terminal window by pressing Ctrl + Alt + T.
- Enter the command ping -c 10 hostname.
- Replace hostname with the IP address or domain name you want to ping.
- The Terminal will display the ping results, including the round-trip time for each packet.

Que: Understand basic firewall configuration using FIREWALL-CMD.

Ans:

- Check the firewall's status: Use the `sudo firewall-cmd --state` command.
- Check the firewall's configuration: Use the `firewall-cmd` command to interact with the `firewalld` configuration.
- Open or close a port: Use the `firewall-cmd --zone=public --add-port=8083/tcp --permanent` command to open a port.
- Reload the firewall configuration: Use the `firewall-cmd --reload` command.

Que: add ssh services in firewall.

Ans: To add SSH services in a firewall,

- Log into SSH
- Check if the application port is defined as a service
- Whitelist the service permanently in the current zone
- Reload `Firewalld` to apply changes

Que: Graphically manage the firewall.

Ans:

- The Security Level Configuration Tool (`system-config-securitylevel`) is a graphical tool in Red Hat Enterprise

Linux (RHEL) that can be used to configure a firewall. It uses a control panel interface to create broad iptables rules for a general-purpose firewall.

- To install the firewall-config GUI configuration tool:
 - ✓ Open the Software Sources application by using the Super key and typing Software
 - ✓ Type firewall in the search box
 - ✓ Select the Firewall item from the search results
 - ✓ Click the Install button

Que: What is selinux security.

Ans: Security-Enhanced Linux (SELinux) is a security module in the Linux kernel that adds an extra layer of security to Red Hat Enterprise Linux (RHEL) and CentOS.

Que: How to set static IP in linux?

Ans: To set a static IP address in linux, we can use the command line and also GUI tool.

- **Command-line:** use the ifconfig command to set the IP address. We can also use the nmcli command line tool to modify the network connection.
- **GUI tool:** Use the network manager text user interface to configure your network interface.